

VIKASH MADDI

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EDUCATION

Indian Institute of Technology Madras, Chennai, India

July 2018 – May 2022

Bachelor of Technology in Mechanical Engineering

8.31/10

PROFESSIONAL EXPERIENCE

VMock, Full Stack Developer

August 2022 – Present

Agentic AI Workloads in Production

- Built and operate a production LLM agent used by 5,000+ users, designed as a planning-first orchestration loop over a structured tool registry with Pydantic-validated schemas as guardrails at every tool boundary.
- Authored an MCP server and HTTP proxy that expose internal tools and data to agents safely, with schema-validated boundaries as the permission layer and no client-side changes required.
- Designed the agent tool-use framework over a structured registry, with retries, timeouts and rate-limit handling around model APIs, and guardrails enforced by schema validation rather than prompt instructions.

LLM Observability, Evaluation and Cost Control

- Stood up LLM observability with Arize Phoenix, instrumenting span-level traces across the agent lifecycle so failures are attributable to the model, the prompt, the tool or the harness.
- Track token, cost and latency per session as first-class platform metrics, and built a context-management layer with prompt caching that cut inference cost on long-running sessions.
- Built evaluation harnesses with LLM-as-judge graders and regression suites over representative cases, so model and prompt changes are gated before rollout.

Platform, Deployment and Reliability

- Operate containerized services on Docker and Kubernetes across AWS (S3, SQS), with CI/CD pipelines from repository to production and Git-based review.
- Built self-serve internal tooling and admin dashboards that automated user onboarding, environment and tenant provisioning, removing 30% of manual operational work and accelerating go-live cycles.
- Led a zero-downtime cross-datacenter migration of 90k+ accounts across US and UK environments with dual-write, hash-based reconciliation to prove parity and staged rollout with rollback at every stage, at 99.9% data integrity.

Python, Data and Retrieval

- Cut API response latency from 400ms to 30ms (13x) by reading query plans, adding targeted indexes, introducing Redis caching and refining service boundaries.
- Built RAG-style retrieval over Elasticsearch with embeddings, chunking, similarity tuning and relevance evaluation.

Karomi Technologies, Deep Learning Intern

May 2021 – August 2021

- Trained Faster R-CNN, RetinaNet and EfficientDet models in PyTorch with semi-supervised learning, reaching mAP of 96% for tables and 95% for symbols.

PROJECTS

Context Management Proxy and MCP Server | *Python, MCP, Agent Infrastructure*

2026

- Implemented relevance-based eviction and prefix-stable prompt caching, then an eval harness measuring cost per completed task before and after compression, so the improvement was proven rather than assumed.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, JavaScript, Bash, C++, PHP, Java

Agentic AI Stack: LangGraph, MCP (authoring and hosting servers), Tool-Use Frameworks, Agent Orchestration, Guardrails, OpenAI/Anthropic APIs, Context Management, Prompt Caching, Claude Code, Cursor

AI Observability & Evaluation: Arize Phoenix, Span-Level Tracing, LLM-as-Judge Graders, Regression Suites, Token/Cost/Latency Tracking

Backend & APIs: FastAPI, Flask, Laravel, REST APIs, Microservices, asyncio, Pydantic, API Versioning

Frontend: React, Redux, Tailwind, HTML5, CSS3

Containers & Cloud: Kubernetes, Docker, AWS (S3, SQS, IAM basics), Cloudflare Workers, Linux, Deployment Automation, Rollback Strategy

SCHOLASTIC ACHIEVEMENTS

- Secured a global rank of 38 in the July Long Challenge on the CodeChef platform.
- Secured an all-India open category rank of 1308 in JEE-ADVANCED 2018 among 1.8 million.